

Omer Pinto

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Summary & Current Focus

- Senior Software Engineer with **15+ years** building **backend systems, database internals, & distributed platforms**.
- Currently developing **Python** microservices and orchestration systems at BigArch while actively expanding into **AI Engineering**.
- Develop **agentic workflows** using **LangGraph / LangChain** through personal projects & professional AI engineering coursework.
- Strong foundation in **algorithms, performance tuning, & systems architecture** across C++, Python, Java, & cloud platforms.
- Seeking **AI Engineering roles** focused on intelligent products and **LLM-powered applications**.

Experience

BigArch

SENIOR SW ENGINEER

2024 - Present

- Backend development of a large-scale **graph orchestration platform** for HPC with **Python, MongoDB, RabbitMQ & microservices**.
- Architect and deploy containerized services in private-cloud environment using **Kubernetes** and **Docker**.
- Achieved **60% performance gains** through system-wide optimization, transforming strained system into stable, scalable platform.
- Lead system-wide architectural refactors executed with **zero downtime** and full service continuity.
- **AI Initiatives:** Evaluated multiple LLMs in air-gapped system. Built custom shim server enabling LLM tool integration.
- Drive AI-assisted development adoption through practical MVPs & team enablement (More on AI work & self-study in Projects section)

Microsoft

SENIOR SW ENGINEER

2023 - 2024

- Engineer in the **vulnerability management** group, developing components in the endpoint security pipeline.
- Built **C# microservices** and data pipelines using Azure services (Blobs, Queues, CosmosDB, KustoDB).

Yahoo

PRINCIPAL SW ENGINEER

2022 - 2023

- Contributor to the **native ads** platform, developing **Java** services and **Hadoop/Spark** ETL pipelines.
- Partnered with research team to implement **ML/DL-driven ad-scoring algorithms** for multi-variable ranking system.
- Managed codebase migration and team handoff following **project cancellation**.

qSpark

SENIOR SW ENGINEER

2021 - 2022

- Delivered ultra-low-latency trading infrastructure. Implemented exchange protocols (execution & market data) using modern C++ with template meta-programming.
- Designed scalable subsystem enabling 3rd-party trading algorithms integration.
- Handled production issues with global exchanges and conducted market-structure research (SEC/FINRA rules).

Toga Networks

HANDS-ON ARCHITECT

2019 - 2021

- Worked across multiple projects in the **cloud and IoT R&D** group, driving architecture and implementation.
- Performed **market and technology research** and delivered high-level designs for innovative systems.
- Led research & design of semantic analytics engine for IoT data, including reasoning capabilities across data model and query layers.
- Developed a log-analysis algorithm using **exponential backoff**, improving system performance.

SiSense

SENIOR ENGINEER, TECH LEAD & ARCHITECT, CTO OFFICE

2014 - 2018

- **Hand-picked by CTO** for his own private team researching next-gen analytics & developing features on open-source DB project.
- Architected and built cross-platform **ETL query engine** from scratch as sole developer; recruited and led team as tech lead.
- Engine optimized for low RAM footprint, 100% multicore CPU utilization (any scale-up) and distributed execution (scale-out), implemented using C++14/17 template meta-programming.
- Implemented core DB features and system optimizations using multicore algorithms and cache-aware data structures.
- Designed thread management solution improving query engine performance by order of magnitude.
- Optimized query operations (join, group) on multi-billion record datasets
- Implemented statistical algorithms from academic papers.
- Contributed by implementing innovative complex DB compiler internals and execution engine features.

Early years - Allot Communications, ZRRO Technologies

FIRMWARE / EMBEDDED ENGINEER

2011-2014

- Design & implementation of **real-time algorithms & data structures** in a resource-restricted, **performance-critical** environment.
- Network protocols, coding of algorithms & systems in **C & C++** using extreme **memory/cpu optimizations** on **Linux** os.

Technical Skills

- **Modern C++ (11, 14, 17, 20):** Extensive experience with the language, including **advanced template metaprogramming** techniques, used for extreme compile-time optimizations in a practical way.
- **Python & Backend Development:** Production experience with microservices architecture, distributed systems (MongoDB, RabbitMQ, Kafka). RESTful APIs, async programming, performance optimization. Docker, Kubernetes for containerized deployments.
- **AI & LLM Systems:** Hands-on development of multi-agent agentic frameworks (LangGraph, LangChain, openAI agents SDK) with MCP protocol integration. Ongoing learning through coursework & hands-on project development.
- **Database Systems & Internals:** Deep expertise in DB internals (query parsers, execution engines, optimizers, caching layers). Designed and built ETL engine from scratch. Working knowledge of PostgreSQL, MongoDB, ClickHouse. Query language DSL design.
- **Additional Languages:** Java (Spring, Maven, Hadoop/Spark), C# (Azure microservices).
- **Cloud technologies:** Azure services (Blobs, Queues, CosmosDB, KustoDB), AWS (EC2, S3).
- **Distributed & graph Systems:** implement Hadoop/Spark pipelines, architect features on distributed graph orchestration platforms.
- **Development Practices:** Git, CI/CD (Jenkins, ADO), Agile/Scrum, unit testing, system design, performance profiling.

Education

The Open University of Israel

Ra'anana, Israel

B.SC IN MATHEMATICS AND COMPUTER SCIENCE

2007 - 2010

- Graduated **summa cum laude** with GPA of **97/100**.
- Awarded place on University President's Excellence List (annual GPA ≥ 95) for four consecutive years.
- Achievement recognized in university educational portal [article](#).

Independent Work & Technical Writing

AI Projects, Udemy AI Learning & Technical Writing

- **LangGraphWrappers:** Infrastructure layer for building LangGraph agents with reusable components (nodes, edges, tools, structured outputs). Demonstrates systems-level thinking applied to agent development frameworks. [GitHub](#).
- **StocksMarketRecommender:** Multi-agent workflow with Yahoo Finance MCP integration. Orchestrates multiple LangGraph flows using LangGraphWrappers library. [GitHub](#).
- **SoccerSmartBet (ongoing):** Real-time betting analysis system comparing AI vs. human strategies. Demonstrates high level architecture of complex independent agentic workflows using LangGraph infrastructure. [GitHub](#).
- Completed **AI Engineer Agentic Track: The Complete Agent & MCP Course** (Udemy, 2025) - comprehensive training in AI agents, multi-agent systems, MCP protocol, prompt engineering, and AI systems design.
- Published technical series on database implementation and C++ techniques on LinkedIn, republished by [EuropeClouds](#) (2020-2021). Created [C++ TMP Tutorial](#) referenced in articles.